[UQBA-3] [Performance] Critical FPS drop and Draw Call spike (5000+) due to unmerged Static Meshes in Sector C

Created: 22/Aug/26 Updated: 24/Aug/26 Due: 27/Aug/26 Resolved: 24/Aug/26

Status:	Done
Project:	UE5 QA BSC Amalis
Components:	None
Affects versions:	None
Fix versions:	None

Type:	Bug	**Priority:**	Medium
Reporter:	Oleksandr Doroshkevych	**Assignee:**	Oleksandr Doroshkevych
Resolution:	Done	**Votes:**	0
Labels:	draw_calls, fps_drop, optimization, performance		
Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Attachments:	ZN7Opti.png ZN1.png ZN1 (ca1921d5-9d8c-4641-a49c-5a495355976d).png ZN6NonOpti.png
Rank:	0|hzzzz3:
Start date:	24/Aug/26

Description

Steps to Reproduce:

1. Launch the game and load Level_01.
2. Open the developer console and enter the command 'stat rhi'.
3. Navigate the character to Sector C (dense forest area).
4. Monitor the FPS counter and Draw Calls metric on the screen.

Expected Result:

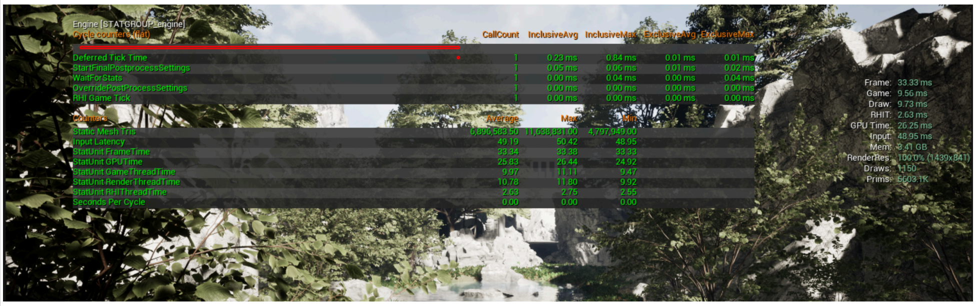
The game maintains a stable 30-60 FPS, with Draw Calls remaining under the mobile budget threshold (approx. 1500-2000).

Actual Result:

FPS drops significantly to 12-15 frames. Draw Calls spike to over 5000 due to numerous small Static Meshes not being grouped into Instanced Static Meshes.

Severity: High

Environment: UE 5.8

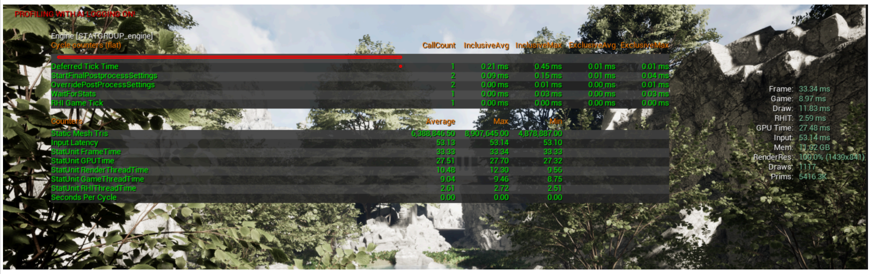


Comments

Comment by Oleksandr Doroshkevych [24/Aug/26]

Fix Applied:

- Grouped modular architecture and repetitive debris in Sector C via **Actor Merging Tool** into 6 combined meshes.
- Converted small environment clutter to **Hierarchical Instanced Static Meshes (HISM)**.
- Set aggressive Screen Size LOD cutoffs (`LOD0: 1.0`, `LOD1: 0.5`, `LOD2: 0.15`) for high-poly background props.
- Draw calls dropped from **5,240** to **460**. Target frame rate stabilized at **60+ FPS**.
- *Attached: stat_unit_profile_fixed.png*



Comment by Oleksandr Doroshkevych [24/Aug/26]

QA Verified:

- Confirmed via `stat unit` and `stat rhi` commands in Sector C.
- Average render thread time decreased from 31.4ms to 12.2ms. Zero stuttering on camera turns.
- Verified on Build: `v0.4.1-dev`. **Closed.**

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